

Reality-First Quantum Gravity

How BM, AS, CDT and CST fit together in plain English

Purpose

This document explains the revised synthesis without requiring a technical background.

The central picture is:

The Stage is what exists around the physical system.

The Dancer is what is physically definite.

The Dance is what makes everything change.

The four frameworks are **not** adopted wholesale. Each contributes a mathematical structure for a particular job, and each assumption is treated as a hypothesis that must survive the Reality-First tests.

The working architecture is:

Causal structure → geometric histories → macroscopic spacetime ↔ matter → actual configuration

with

renormalization flow

organizing how the description changes from very small scales to large scales.

1. Why quantum gravity is needed

Physics currently has two enormously successful descriptions that do not fit together cleanly at the deepest level.

General relativity describes gravity as the geometry of spacetime.

Quantum theory describes matter and its possible states using quantum mechanics and quantum field theory.

The difficulty appears when spacetime itself must behave quantum mechanically—for example near the Planck scale, inside extreme gravitational environments, or in the very early universe.

The basic quantum-gravity question is therefore:

What is the Stage at the deepest level, and how does the ordinary spacetime we observe emerge from it?

The Reality-First program does not assume that the answer must already be one existing theory.

Instead it asks:

What must exist? → What mathematics can represent it? → What prediction follows?

Only then:

Does reality distinguish that prediction from a null model?

2. The Stage, Dancer and Dance

Imagine a theater.

The **Stage** is the spacetime arena.

The **Dancer** is the physical system occupying that arena.

The **Dance** is the dynamics: the rules governing how the Stage and Dancer change.

This metaphor separates jobs that are often mixed together.

Physical idea	Plain-English meaning	Mathematical language
Stage	spacetime, geometry, causal structure	causal sets, triangulations, metrics, differential geometry
Dancer	matter, states, observables, definite configuration	operator algebras, Hilbert spaces, fields, configurations
Dance	change, interaction, scale dependence	actions, differential equations, path integrals, RG flow
Constitution	things that remain stable across descriptions	spectra, correlations, symmetries, critical exponents

The key Reality-First warning is:

A mathematical representation is not automatically the physical thing it represents.

A lattice is not automatically space.

A spectrum is not automatically a particle.

A causal order is not automatically a metric.

An RG fixed point is not automatically a microscopic object.

A probability formula is not automatically a physical mechanism.

3. Where Causal Set Theory enters

Causal Set Theory, or **CST**, begins with a very simple idea:

Before talking about distances, coordinates or a smooth metric, perhaps the most basic fact is simply that some events can causally precede others.

A causal set can be represented as

$$C = (E, <).$$

Here:

- (E) is a set of events.
- $(x \prec y)$ means that event (x) can causally precede event (y) .

The relation is ordered and locally finite.

In plain English:

The universe starts, at least as a candidate description, with events and causal precedence rather than a ready-made smooth spacetime.

This gives us something important that geometry by itself does not have to provide first:

causal order.

It also gives a natural notion of discrete counting. If an interval contains many causal events, that count can play the role of a discrete volume measure.

So CST contributes:

causal precedence + local finiteness + counting

What CST does not automatically give us

A causal order does not by itself tell us:

- the exact distances between events,
- the metric tensor,
- a unique Lorentzian manifold,
- the observed four-dimensional large-scale world,
- a complete quantum dynamics.

This is precisely where the Reality-First discipline matters.

We therefore do **not** say:

“CST has already derived spacetime.”

We say:

“CST gives us a candidate pre-geometric causal substrate. We now have to show that the observed geometric Stage can actually be reconstructed from it.”

That is the first major bridge.

causal substrate $\longrightarrow$ spacetime
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This bridge is **open**, not assumed.

4. Where CDT enters

Causal Dynamical Triangulations, or **CDT**, asks a different question.

Suppose we want to calculate quantum histories of geometry without simply assuming one smooth classical geometry.

A practical way to do that is to represent geometries using causal simplicial structures and sum over allowed histories.

Think of it as building a huge catalog of possible causal geometric histories.

The schematic form is

$$Z = \sum_T \frac{1}{C_T} e^{iS[T]}.$$

You do not need the equation to understand the idea:

Instead of choosing one geometry, we consider many allowed causal geometries and let the quantum sum determine which large-scale behavior dominates.

This is the important contribution from CDT.

It supplies a **controlled dynamical laboratory for geometry**.

But the Reality-First interpretation is deliberately narrower than the strongest possible interpretation of CDT.

We do **not** automatically say:

“The universe is literally made of simplices.”

Instead:

“Causal triangulations are a mathematical representation through which we can define and test a nonperturbative geometric ensemble.”

That distinction matters.

A simplex is a representation.

The physical question is whether the ensemble produces the right large-scale observables.

5. CST and CDT therefore do different jobs

This is one of the most important parts of the revised synthesis.

CST and CDT are not being treated as two competing answers to exactly the same question.

Their mathematical jobs are different.

Framework	Main question	What we retain
CST	What can exist before metric geometry?	causal order, local finiteness, discrete cardinality
CDT	How can causal geometries have a quantum dynamics?	causal geometric histories, path sum, phase structure

The desired bridge is:

$CST \longleftrightarrow CDT \longrightarrow \text{macroscopic Stage}$

But the arrows are claims to be **tested**, not definitions.

We need to know whether the causal information in CST can be represented in a suitable CDT ensemble and whether both descriptions converge on the same physical observables.

That is much stronger than simply saying that the two theories “look compatible.”

6. The Stage at large scales

The world we actually observe looks like a smooth Lorentzian spacetime.

So the candidate microscopic description must eventually reproduce something like

$(M, g_{\mu\nu}).$

Here:

- (M) is the spacetime manifold.
- $(g_{\mu\nu})$ is the metric describing geometry.

At ordinary scales, the Stage has familiar properties:

- distances,
- durations,
- causal cones,
- gravitational curvature,
- wave propagation.

The Reality-First requirement is therefore:

microscopic description $\longrightarrow$ stable macroscopic Lorentzian spacetime

And “stable” matters.

Seeing a four-dimensional pattern in a simulation is not enough.

A genuine continuum limit requires controlled scaling as the microscopic regulator is removed. In schematic form,

$$\frac{\xi}{a} \rightarrow \infty,$$

where (a) is the microscopic scale and (ξ) is the relevant correlation length.

Plain English:

The large structures must become much larger than the microscopic bookkeeping scale, so that the macroscopic physics stops depending on the details of the bookkeeping.

That is the real continuum test.

7. Where Asymptotic Safety enters

Now we reach **Asymptotic Safety**, or **AS**.

AS addresses a different problem:

How can the laws of physics remain mathematically controlled as we move between very small and very large scales?

Instead of assuming that the same simple equation works unchanged at every scale, AS studies an effective description

$$\Gamma_k$$

that changes with a scale (k).

Think of (k) as the resolution of our description.

At high (k), we look at very short distances.

At low (k), we describe large-distance physics.

The RG flow is schematically

$$k \frac{dg_i}{dk} = \beta_i(g).$$

A fixed point satisfies

$$\beta_i(g^*) = 0.$$

In plain English:

A fixed point is a scale regime where the essential dimensionless couplings stop running in the appropriate sense.

This gives us a candidate mechanism for **cross-scale organization**.

AS therefore contributes:

UV $\rightarrow$ IR scale flow

not the microscopic Stage itself.

8. CDT and AS are not the same thing

This distinction is essential.

CDT asks:

What happens when we sum over causal geometric histories?

AS asks:

How does the effective description of the physics change with scale?

So one is a **geometric construction / laboratory** and the other is a **cross-scale flow framework**.

The synthesis therefore tests:

CDT critical behavior $\leftrightarrow$ AS universality

rather than simply declaring

$$\text{CDT} = \text{AS}.$$

A shared exponent, scaling law or dimensionless observable can provide evidence of a common universality class.

But agreement must survive:

- regulator changes,
- discretization changes,
- finite-size effects,
- observable matching.

That is what makes it a physics test rather than an analogy.

9. Spectral dimension: a useful thermometer

One of the useful quantities in quantum-gravity work is the **spectral dimension**.

Imagine releasing a tiny random walker into the geometry.

Ask:

How quickly does the walker return to where it started?

From the return probability ($P(\sigma)$), define

$$D_S(\sigma) = -2 \frac{d \ln P(\sigma)}{d \ln \sigma}.$$

In ordinary flat (d)-dimensional space,

$$D_S = d.$$

So (D_S) behaves like a kind of thermometer for the effective dimensional behavior of the geometry.

A famous class of quantum-gravity results finds scale dependence in which the short-distance and long-distance effective dimensions can differ.

The Reality-First discipline says:

Spectral dimension is an observable of the geometry. It is not automatically the literal number of physical dimensions.

And:

$$\sigma \propto k^{-2}$$

as a definition.

Diffusion time and RG scale are different mathematical concepts. If they are related, that relation has to be established.

10. Now introduce the Dancer: matter

So far we have mainly discussed the Stage.

But a universe is not just geometry.

There is matter.

This is where the Dancer enters.

The Dancer is represented using structures such as:

$$(A, H, \Phi)$$

where, roughly:

- $(\mathcal{A})$ describes observables,
- $(\mathcal{H})$ describes quantum states,
- (Φ) describes matter fields.

The exact particle content and gauge structure are not automatically supplied by CST or CDT.

That is intentional.

The Reality-First corpus explicitly treats the particle census, gauge group and related Standard Model content as things that must not be silently “derived” just because the geometry happens to exist.

So:

$$\boxed{\text{Stage} \boxplus \text{Dancer}}$$

The Stage provides the physical arena.

The Dancer supplies the matter degrees of freedom.

11. But the Dancer cannot be a spectator

This is a critical test.

If we simply put matter on a background geometry and matter never changes the geometry, then the coupling is incomplete.

The correct physical picture is two-way:

$$\boxed{\text{geometry} \leftrightarrow \text{matter}}$$

The combined action is schematically

$$S[g, \Phi] = S_{\text{geometry}}[g] + S_{\text{matter}}[g, \Phi].$$

The matter part depends on the geometry.

The effective action therefore has the form

$$\Gamma_k[g, \Phi].$$

The test is whether changing the matter sector actually changes geometric observables.

Schematically,

$$\frac{\delta \Gamma}{\delta g} \neq 0.$$

Plain English:

Matter must actually push back on the Stage.

This is why the revised synthesis introduces an explicit matter-backreaction gate.

It prevents us from claiming a successful unification when the two sectors merely sit beside one another.

12. Where Bohmian Mechanics enters

Now we reach **Bohmian Mechanics**, or **BM**.

BM addresses a different question:

What, physically, is definite?

Standard quantum mechanics gives a wavefunction or quantum state.

BM adds a definite configuration.

In the simplest textbook case, this is represented by particle positions

$$Q(t).$$

The configuration is not merely “one possible outcome” in the interpretation.

It is the actual configuration.

The quantum state evolves according to the usual wave equation, while the actual configuration evolves according to a guidance law.

Schematically:

$$\dot{Q} = \frac{j}{\rho}.$$

Plain English:

The wave describes the quantum structure, while the guidance law tells the actual configuration how to move.

This gives us a genuine Dancer.

13. But we do not simply import textbook BM

This is another major Reality-First correction.

Traditional BM often begins with a fixed configuration space such as

$$\mathbb{R}^{3N}.$$

That makes sense for (N) nonrelativistic particles in ordinary space.

But our synthesis is trying to explain spacetime itself.

So we cannot simply say:

“Here is $(\mathbb{R}^{3N})$; now put quantum gravity underneath it.”

That would quietly assume the very structure we are trying to recover.

Instead the question becomes:

Can an actual configuration space be constructed from the recovered Stage and matter sector?

We therefore write more generally:

$$Q(t) \in Q[M, \Phi].$$

Plain English:

The actual Dancer must live in a configuration space that is justified by the physical theory we have actually constructed.

This is the BM interface.

It is a construction problem, not an assumption.

14. What the BM mathematics gives us

Suppose the quantum state is

$$\Psi = Re^{iS/\hbar}.$$

BM identifies a current (j) associated with the quantum state.

The actual configuration follows

$$\dot{Q} = \frac{j}{\rho}.$$

The same dynamics produce a continuity equation for the probability density:

$$\partial_t \rho + \nabla \cdot (\rho v) = 0.$$

The quantum state itself satisfies the corresponding conservation equation for

$$|\Psi|^2.$$

Therefore, when

$$\rho = |\Psi|^2,$$

the two distributions evolve together.

This is called **equivariance**.

In plain English:

Once the actual configurations start with the quantum distribution, the BM dynamics preserve that relationship.

That is a real mathematical result under the required assumptions.

But it is important not to overstate it.

Equivariance does **not** by itself explain why the universe initially had

$$\rho = |\Psi|^2.$$

15. The Born rule problem

This is one of the Reality-First “silences.”

The observed quantum rule says that measurement frequencies follow the familiar squared-amplitude weighting:

$$W(E) = \langle \Psi, P_E \Psi \rangle.$$

The program does not simply declare:

“BM plus chaos proves the Born rule.”

Instead we separate two questions.

Selector

Why is

$$|\Psi|^2$$

the physically relevant measure at all?

Equilibrator

Why do actual configurations move toward that measure if they start away from it?

These are different problems.

Valentini-style subquantum relaxation provides a possible **equilibrator**.

Define

$$H = \int \rho \ln \frac{\rho}{|\Psi|^2}.$$

Under suitable coarse-graining and mixing assumptions, a relaxation program can produce a decrease of coarse-grained (H).

Plain English:

A sufficiently mixing dynamical system can wash out deviations from quantum equilibrium.

That is interesting.

But it is not the same thing as proving that quantum equilibrium is the unique fundamental measure.

So the Reality-First position is:

H-theorem $\nRightarrow$ complete Born-rule derivation
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The Born interface remains a real research gate.

16. The whole system now fits together

We can now tell the complete story.

Step 1 — Causal substrate

CST gives the most economical candidate for what survives before a smooth metric:

$$C = (E, <)$$

Events exist in causal order.

Step 2 — Geometric dynamics

CDT supplies a way to construct and sum over causal geometric histories:

$$C \rightarrow T \rightarrow Z_{\text{geom}}$$

The triangulations are the mathematical machinery of the calculation, not automatically the final ontology.

Step 3 — Cross-scale flow

AS organizes how the effective description changes with scale:

$$\Gamma_k : UV \rightarrow IR$$

The important question is whether the discrete geometric sector and RG flow exhibit the same physical universality.

Step 4 — Emergent Stage

At sufficiently large scales, the successful sector must reproduce a stable Lorentzian spacetime:

$$T, C \rightarrow (M, g_{\mu\nu})$$

with the observed low-energy geometric behavior.

Step 5 — Dancer appears

Matter fields and quantum states live on that recovered physical structure:

$$(M, g) \leftrightarrow (A, H, \Phi)$$

The coupling is two-way.

Step 6 — Actual configuration

BM asks which configuration is actually realized:

$$Q_{\text{actual}}(t) \in Q[M, \Phi]$$

and evolves it using a guidance law.

Step 7 — Frequencies

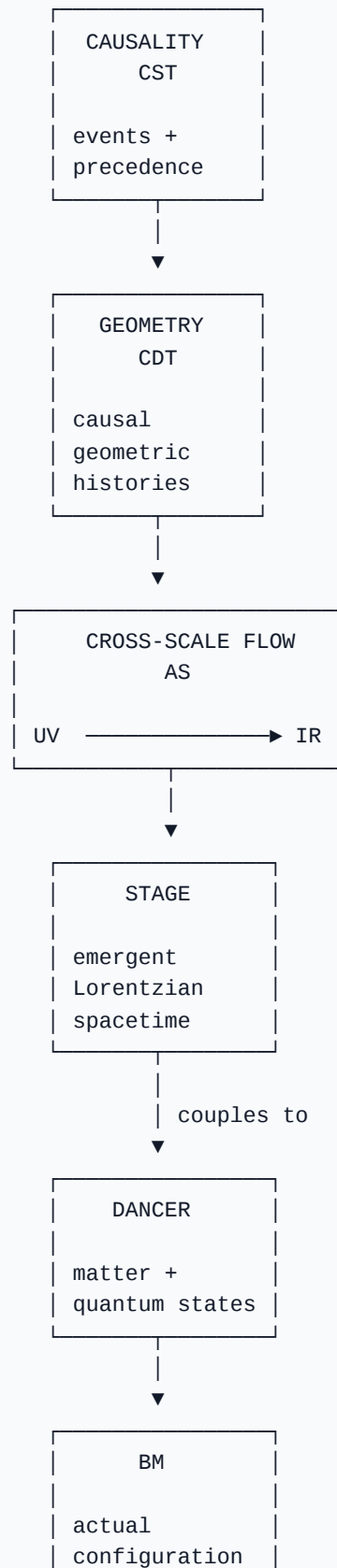
The final observational layer asks whether the actual configurations reproduce empirical quantum frequencies:

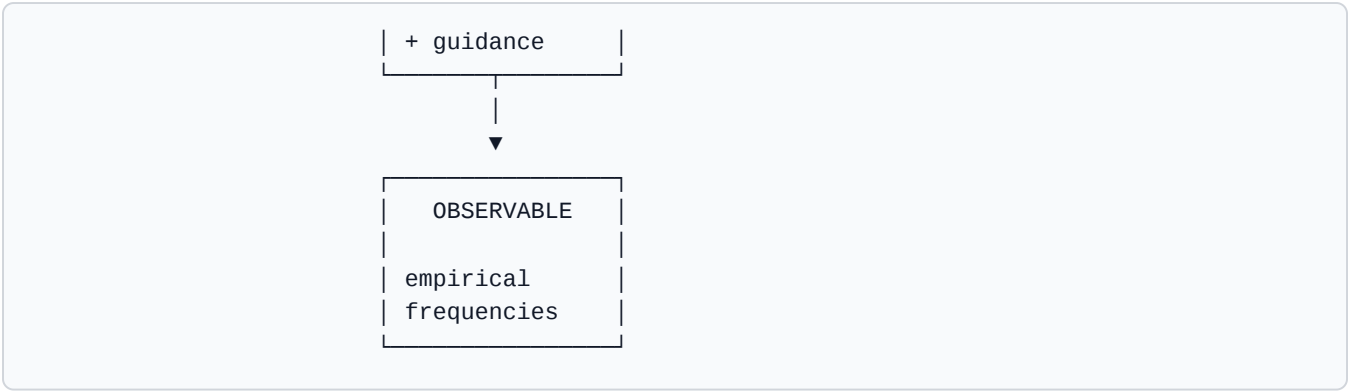
$$Q_{\text{actual}} \rightarrow W(E) \stackrel{?}{=} \langle \Psi, P_E \Psi \rangle$$

That is the endpoint of the chain.

17. The entire picture in one diagram

REALITY-FIRST QUANTUM GRAVITY





The important point is that **the four theories do not all occupy the same box**.

They are pieces of a larger construction.

18. Why this is called Reality-First

The corpus provides a strict discipline for deciding whether each mathematical move deserves physical meaning.

The rule is:

mathematical structure → physical mechanism → prediction → null test

That means every major step must answer four questions:

Question	Meaning
What mathematics is being used?	What is the formal structure?
What physical job does it perform?	Why is that mathematics here?
What new observable follows?	What could reality actually distinguish?
What could kill it?	What result would make us reject it?

This protects the synthesis from several recurring errors.

19. The mathematical compliance rules

Rule 1 — Arithmetic counts; it does not automatically create geometry

The arithmetic audits found strong obstructions for specific rigid classes of discrete dynamics.

But the conclusion is deliberately scoped:

specific arithmetic obstruction $\nVdash$ universal impossibility of discreteness

Therefore CST is not killed merely because some arithmetic substrates fail.

Rule 2 — Causality is not yet geometry

A causal order can tell us who can precede whom.

It does not automatically provide a complete metric.

Therefore:

$$C = (E, <) \rightarrow (M, g)$$

is a reconstruction problem.

That keeps CST honest.

Rule 3 — A triangulation is not automatically the universe

CDT triangulations are retained because they give us a calculational handle on causal geometric histories.

We do not promote each simplex to an elementary physical atom merely because the simulation uses one.

Thus:

$$\text{discrete regulator/representation} \not\equiv \text{physical ontology}$$

unless an independent physical test demands that interpretation.

Rule 4 — RG flow is not microscopic ontology

AS gives a powerful language for scale dependence.

But

$$\Gamma_k$$

is an effective object.

It should not automatically be treated as the microscopic furniture of reality.

Likewise, a fixed point is a mathematical fixed point of the chosen RG description—not automatically a literal physical location in spacetime.

Rule 5 — Matter must be dynamical, not decorative

A synthesis in which matter merely sits on geometry has not yet demonstrated unification.

The matter sector must feed back into the Stage:

$$\delta\Gamma/\delta g \not\equiv 0$$

in an appropriate tested regime.

That is a genuine joint between Dancer and Stage.

Rule 6 — BM must earn its configuration space

BM gives a powerful answer to the question:

What is actually definite?

But a quantum-gravity version cannot simply assume ordinary

$$\mathbb{R}^{3N}.$$

The required configuration space must emerge from the physical framework actually constructed.

Rule 7 — Equivariance is not the Born rule

BM can preserve quantum equilibrium if quantum equilibrium is already present.

That is not the same as deriving why the universe is in quantum equilibrium.

A relaxation mechanism may help with the second question, but the logical distinction remains:

equilibrium preservation $\nleftrightarrow$ equilibrium selection

Rule 8 — Invariants compare theories; they do not generate the world

Spectra, correlations, critical exponents and topological quantities are valuable because they can survive changes in representation.

But they are primarily **comparison tools** and physical observables.

We should not reverse the logic and say:

“Because an invariant exists mathematically, it must be the mechanism generating reality.”

20. What the synthesis is actually claiming

The synthesis is **not** currently claiming:

CST has proved the universe is fundamentally a causal set.

It is not claiming:

CDT has proved the continuum limit of quantum gravity.

It is not claiming:

AS has proved a unique ultraviolet fixed point.

It is not claiming:

BM has derived the Born rule.

It is not claiming:

Matter has been derived from geometry.

Instead, it claims something narrower and more useful:

There is a coherent chain of mathematical jobs in which causal structure, geometric dynamics, scale flow, matter, and actual configurations can be tested as parts of one architecture.

The architecture is:



with

$$\boxed{\Gamma_k}$$

organizing the cross-scale description.

21. What would count as success?

The program survives only if the interfaces survive.

CST $\rightarrow$ Stage

Can causal information be reconstructed into stable macroscopic geometry without secretly importing the continuum that was supposed to emerge?

CST $\leftrightarrow$ CDT

Do the two causal descriptions agree on common observables?

CDT → continuum

Does a genuine continuum sector exist, with the right macroscopic causal and geometric behavior?

CDT ↔ AS

Do the discrete geometric critical properties match the universality structure predicted by RG analysis?

Stage ↔ matter

Can matter be coupled consistently?

Matter → Stage

Does matter actually produce geometric backreaction?

Stage → BM

Does the recovered physical system admit a well-defined actual configuration space?

BM → observation

Do actual configurations produce the observed empirical measure?

These are not philosophical questions.

They are **research gates**.

22. The final mental model

A nontechnical reader can keep the whole program in one picture.

The Stage

At the deepest level, we do not assume a finished smooth spacetime.

We start with a candidate causal structure, test a dynamical geometric realization, and ask whether a smooth four-dimensional spacetime emerges at large scales.

The Dance

The universe does not merely contain a geometry.

There must be laws governing change.

CDT gives a way to calculate geometric histories.

AS gives a way to track how effective laws change with scale.

Together they provide candidate pieces of the Dance—but neither is assumed to be the whole Dance.

The Dancer

Matter is not reducible to the geometry merely because both are described mathematically.

The matter sector needs its own state structure.

BM then adds something philosophically and physically distinctive:

there is an actual configuration.

The quantum state tells us the structure of the possibilities; the actual configuration tells us what is physically realized.

The final bridge

The final bridge from theory to experiment is the hardest one.

The theory must explain why the distribution of actual configurations corresponds to the quantum frequencies we observe.

That is where the Born-measure problem remains.

23. The whole synthesis in one sentence

CST supplies a candidate causal beginning; CDT supplies a calculable ensemble of causal geometries; AS supplies a candidate law for how the effective description changes across scales; the emergent geometry becomes the Stage; matter supplies the Dancer; BM supplies an actual configuration and its guidance law; and the final test is whether this entire chain reproduces the empirical quantum world without importing the answers it was supposed to explain.

That is the Reality-First version of the idea.

The theories are not the answer.

They are **mathematical applicants for different physical jobs**.

Reality decides which parts survive.